PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE

MOISTENING MILL POST

润磨机岗位

MTI-CHP-CPM-SOP-010_Rev. 000

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1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the moistening mill post under complete closure.

此程序涵盖操作员对完全关闭下的润磨机岗位进行目视操作前检查、启动运行和停车的责任。

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that the moistening mill post is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the moistening mill post and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对润磨机岗位进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定润磨机岗位的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures.MTI-CP-SOP岗位操作规程
- Moistening Mill (4301-MIL-001/002).润磨机 (4301-MIL-001/002)
- Belt Conveyors(4301-BCV-003).胶带机(4301-BCV-003)
- Exhaust Fan(4301-FAN-001).引风机(4301-FAN-001)
- Scrubbing Tower(4301-TOW-001).洗涤塔(4301-TOW-001)
- Circulating Water Pump(4301-PSO-001/002).循环泵(4301-PSO-001/002)

4. PPE 个人保护装备



耳塞　安全帽　眼镜　劳保鞋　手套　工作服



防尘口罩

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5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue 危害	Control Strategy 控制
1. Mechanical injury; 2. Dust; 3. Electric shock; 4. Slippery conditions and a trip to the danger; 5. Asphyxia; 6. Noise; 7. Object falling from high altitudes. 1. 机械伤害； 2. 灰尘； 3. 触电； 4. 湿滑的条件，绊倒危险； 5. 窒息； 6. 噪音； 7. 高空坠物。	1. Install a protective cover on the rotating part of the equipment; 2. Wear dust mask as required; 3. Do not contact the electrical switch with wet hands during operation, Regular inspection of the equipment leakage protection device is complete, effective; 4. Please hold the handrail up and down the stairs; 5. Clear clearance or maintenance operations; 6. Wear the earplugs as required to protect your hearing; 7. Wear labor protection goods and check the safety of stacking items on the operation platform. 1. 设备的转动部位安装防护罩； 2. 按要求佩戴防尘口罩； 3. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效； 4. 上下楼梯请扶好扶手； 5. 清仓或维修作业执行审批程序； 6. 按要求佩戴好耳塞，保护听力； 7. 穿戴好劳保护品，检查各平台的物品堆放安全性

6. Equipment 工具/ 设备

4301-CHE-008/009	Chute溜槽
	Clean the site tools清扫现场工具

7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training; 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site; 3. Check and clean the equipment and the site sanitation; 4. Contact with the process before and after, and give

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№	Stage	Key Points
		clear instructions. 1. 确认经过岗前技术培训； 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患； 3. 检查并打扫设备、现场卫生； 4. 与前后工序取得联系，并发出明确指令。
2	Inspect Moistening Mill (4301-MIL-001/002) 检查润磨机 (4301-MIL-001/002)	1. Check whether the lubrication system is normal: a. Start the lubrication system to cycle the oil to the rotary kiln (4304-KLN-001) system. First, wrap a layer of 100 copper wire mesh at the tubing outlet. Then check the filter screen for impurities for 4 hours after the oil circulation. If the oil quality is good, remove the filter screen and restore the tubing system; b. After the lubrication system is opened, check whether there is abnormal vibration and abnormal sound, whether the oil leakage, whether the oil pressure is stable, whether the oil volume meets the requirements, whether the oil return is normal, whether the electrical system and temperature rise are normal. 2. Check whether the high pressure parts and low pressure parts of the lubricating oil station are normal, and flow rate are normal; 3. Check that the lubrication oil injection of the large tooth ring is good, and the oil nozzle is not blocked; 4. The lubricating oil station requires the inlet temperature of circulating water to be lower than 35℃, the circulating water pressure is 0.2~0.4MPa, the circulating oil output is 16MPa, the output is low pressure 0.4~0.5MPa; 5. Observe whether the oil film of the mill is good from the oil mirror; 6. Check the temperature of the main bearing tile body between 40 ~ 65℃, check the oil level indicator of the main bearing body, check the oil level indicator at the shaft neck, the cooling water, whether the gasket leaks, the abnormal sound and vibration, and the operation of the spherical seat; 7. Check the cleanliness of the oil filter in the main bearing oil supply station, control the oil temperature at < 65℃, check the inlet temperature of the oil cooler, the temperature of cooling water at the entrance and exit and the flow of cooling water are normal;

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№	Stage	Key Points
		8. The tightness state of the cylinder liner installation bolt, whether the bolt hole and entry door leak material; 9. Observe whether the sliding state of the material is normal from the discharge cover and whether there is any abnormal sound; 10. Check the horizontal error of the low speed shaft of the reducer, and whether the transmission shaft has abnormal vibration; 11. Determine the high pressure of the main bearing oil supply station, determine the oil pressure of the reducer, the clutch for the auxiliary transmission device must be completely engaged, and the clutch handle must be kept in the correct position; 12. Before driving, it should be judged whether the rotation direction of the mill meets the work requirements; 13. Check whether the front end of the grinding machine is running normally. 1. 检查润滑油站是否正常： a. 启动润滑油站对系统进行油循环，先在油管出口包扎一层100目的铜丝网，油循环进行4小时后检查滤网上是否有杂质，如油质良好，拆除滤网，恢复油管系统； b. 润滑油站开启后，检查是否有异常振动和异响，是否漏油，油压是否稳定，油量是否符合要求，回油是否正常，电气系统和温升是否正常。 2. 检查润滑油站的高压部分和低压部分是否正常，流量是否正常； 3. 检查大齿圈润滑喷油情况良好，油喷嘴没有堵塞； 4. 润滑油站要求循环水进口温度低于35℃，循环水压力0.2～0.4MPa，循环油输出高压16MPa、输出低压0.4～0.5MPa； 5. 从检查油镜观察润磨机轴瓦油膜情况是否良好； 6. 检查主轴承瓦体温度在40～65℃之间，从主轴承主体上盖的窥视孔检查轴颈处润滑油状态油位计指示，油位计指示，冷却水通水状态，密封垫处是否漏油，有无异常声音和振动，球面座的运转情况； 7. 检查主轴承供油站油过滤器过滤网清洁程度，检查给油温度控制在＜65℃，检查油冷却器的入口温度，出

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№	Stage	Key Points
		入口冷却水的温度以及冷却水的流量是否正常； 8. 筒体衬板安装螺栓的松紧状态，螺栓孔、入孔门是否漏料； 9. 从出料口观察料滑动状态是否正常及有无异常声音； 10. 检查减速机低速轴等的水平误差，传动轴有无异常振动； 11. 确定主轴承供油站高压给油压力，确定减速机给油压力，辅助传动装置用离合器必须完全啮合，离合器手柄必须保持正确位置； 12. 开车前应判明润磨机旋转方向是否符合工作要求； 13. 检查润磨机前端称重皮带秤给料机运转是否正常。
3	Inspect Belt Conveyors(4301-BCV-003) 检查胶带机(4301-BCV-003)	1. Each bearing is well lubrication and closed, the reducer oil level is appropriate, the upper and lower roller lubrication is good, the operation is flexible, whether the wear of the first and tail wheels and the tape are filled with materials, and whether the cleaning device of the first and tail wheels is in good condition; 2. Comprehensively check whether all parts of the belt conveyors are intact, whether the tension of the tape is appropriate, and whether the corresponding tape tightening device and dust-proof facilities are in good condition; 3. Whether the electrical facilities and control system are intact, whether the protective device is firm, and whether the funnel baffle and liner are damaged; 4. Check whether the cushion rollers of the impact load parts are intact; 5. Check the installation of belt sweeper; 6. Whether the installed electrical facilities and signal devices are intact and responsive; 7. Place the field control box in the manual position, press the start button, slowly rotate the potentiometer, and watch whether the running direction of the tape is correct, and whether the belt can run away; 8. After checking and contacting the post, press the start button; 9. During the operation process, check whether the tape machine emergency stop device, belt deviation correction and speed measuring sensor are in good condition. 1. 各部轴承润滑与封闭良好，减速机油位适当，上下托

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№	Stage	Key Points
		滚润滑良好，运转灵活，首、尾轮磨损情况与胶带间是否夹有物料，首、尾轮的清扫装置是否完好； 2. 全面检查胶带机各部分是否完好无损，胶带的张紧度是否合适，相应的胶带拉紧装置及防尘设施是否完好； 3. 电器设施和控制系统是否完好，防护装置是否牢固，漏斗挡板和衬板是否损坏； 4. 检查承受冲击载荷部位的缓冲托辊是否完好； 5. 检查胶带机清扫器安装是否正常； 6. 装设的电器设施及信号装置是否完好和反应灵敏； 7. 将现场控制盒置于手动位置，按一下启动按钮，迟缓旋动电位器，观看胶带运转方向能否正确，皮带是否跑偏； 8. 检查无误并与前后岗位联系后，按下启动按钮； 9. 运转过程中，检查胶带机急停装置、皮带纠偏和测速传感器是否控制灵敏。
4	Inspect Exhaust Fan(4301-FAN-001) 检查引风机(4301-FAN-001)	1. Manual turning 2~3 turn, flexible rotation, no scraping, stuck and other abnormal phenomena; 2. The washing tower (4301-TOW-001) is normal, and the fan inlet butterfly valve (80BF) is closed; 3. Whether the coupling concentricity meets the requirements; 4. Check whether the lubricating oil level of the fan bearing box is normal; 5. Close the power switch, first click (press the operation button and quickly press the stop button) to confirm that the steering is correct, no noise, no vibration and other abnormal conditions; 6. After the fan is started and running, observe the change of the fan sound, check the temperature and noise of the fan bearing and the motor bearing, and check the fan vibration (including the impeller). 1. 手动盘车2～3转，转动灵活，不得有刮蹭、卡住等异常现象； 2. 洗涤塔(4301-TOW-001)出风正常，风机进口蝶阀(80BF)关闭； 3. 联轴器同心度是否符合要求； 4. 检查风机轴承箱润滑油位是否正常；

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№	Stage	Key Points
		5. 合上电源开关，先点动(按运转按钮后快速按停止按钮)确认转向正确、无杂音、无振动等异常情况； 6. 风机启动运行后，观察风机声音的变化，检查风机轴承、电机轴承的温度和噪声，检查风机振动(包括叶轮)。
5	Inspect Scrubbing Tower(4301-TOW-001) 检查洗涤塔(4301-TOW-001)	1. Remove all sundries in the scrubbers and other equipment; 2. Check whether the instruments are complete, and whether the liquid level meter, pressure and thermocouple display are normal; 3. Check whether the liquid circulation pipeline, gas pipeline and valve are in good condition; 4. Open the new water gate valve (50GA) before the tower circulation tank, inject the tower into new water, open the circulation pump (4301-PSO-001 / 002) inlet and outlet butterfly valve (80BF, 80BF), and open the circulation pump (4301-PSO-001 / 002); 5. Observe the spray washing situation of the washing tower, whether the spray amount and spray density meet the requirements; 6. Check whether the spray support ring, fog gate plate, screen separator and nozzle assembly are normal, and the spray remover and nozzle are not blocked; 7. Check whether the packing arrangement, compactness and height of the packing system meet the requirements, and whether the spray liquid dispersion is uniform; 8. Check whether the spray port, access port, sewage outlet, gas outlet, pump outlet, water supply outlet, manhole are normal, whether the connection bolts are fastened, whether leakage or leakage; check whether the tower body leakage or leakage; 9. Empty the inspection water in the washer tower. 1. 清除洗涤塔等设备内的一切杂物； 2. 检查各处仪表是否齐全，液位计、压力和热电偶显示是否正常； 3. 检查各液体管线、气体管路、阀门是否完好； 4. 打开洗涤塔循环槽前新水闸阀(50GA)，将洗涤塔注入新水，打开循环泵(4301-PSO-001/002)出入口蝶阀(80BF,80BF)，开启循环泵(4301-PSO-001/002)；

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		5. 观察洗涤塔喷淋洗涤情况，喷淋量、喷淋密度是否符合要求； 6. 检查喷淋支座环、除雾栅板、丝网除沫器、喷嘴组件是否正常，除沫器和喷嘴有无堵塞现象； 7. 检查填料系统的填料排布、密实度和高度是否符合要求，喷淋液散布是否均匀； 8. 检查喷淋口、检修口、排污口、气体进出口、泵口、补水口、人孔等是否正常，连接螺栓是否紧固，是否漏液或漏气；检查塔体是否漏液或漏气； 9. 放空洗涤塔内的检查用水。
6	Inspect Circulating Water Pump(4301-PSO-001/002) 检查循环泵(4301-PSO-001/002)	1. Check whether the pump inlet pipeline and auxiliary pipeline, flange and valve installation meet the requirements, whether the anchor bolts and ground wire are good, and whether the coupling is installed; 2. Turn the pump shaft 3 times, should be flexible without block; 3. Check whether the lubricating oil level is normal. The oil level should be between 1/2 and 2/3 of the oil lens; 4. Check the sealing condition of the pump is normal; 5. Point to move to confirm the correct steering; 6. Check whether the pump control cabinet voltmeter, signal lights and other instrument indicators are normal; 7. Check whether there is air in the pump unit, if it should be excluded; 8. Check that the pump head is at the level of the scrubbing tower(4301-TOW-001); 9. Check that the conveying pump is in the normal condition,and the outlet valve (80BF) are temporarily closed after opening flexibly; 10. Check the backup pump(80BF) outlet valve closed. 1. 检查泵出入口管线及附属管线，法兰、阀门安装是否符合要求，地脚螺栓及地线是否良好，联轴器及防护罩是否装好； 2. 转动泵轴3圈，应灵活无阻滞； 3. 检查润滑油油位是否正常，油位应在油镜的1/2～2/3 之间； 4. 检查泵的密封状况是否正常；

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		5. 点动确认转向正确； 6. 检查泵控制柜电压表、信号灯等仪表指示是否正常； 7. 检查泵机组是否有空气，如有应予以排除； 8. 检查泵头是否处于洗涤塔(4301-TOW-001)液位下； 9. 检查输送泵处于正常状态，出口阀门(80BF)开启灵活后暂时处于关闭状态； 10. 检查备用泵出口阀(80BF)全关。
7	Check the other Equipment and Facilities 检查其它设备设施	1. Check whether the inlet and outlet pipelines and auxiliary pipelines of the scrubber tower (4301-TOW-001), flange and valve(50GA) installation meet the requirements, and whether the tower body leaks liquid; 2. Check whether the valves (80BF,80BF,80BF,80BF,80BF) of circulating water pump(4301-PSO-001/002) are normal; 3. Check whether the belt chute(4301-CHE-008/009) of the tape machine before the grinder is normal; 4. Check whether the inlet gate valve (65GA) of the cooling water of the grinder lubricating oil station is normal. 1. 检查洗涤塔(4301-TOW-001)出入口管线及附属管线，法兰、阀门(50GA)安装是否符合要求，塔身是否漏液； 2. 检查循环泵(4301-PSO-001/002)各阀门(80BF,80BF,80BF,80BF,80BF)是否正常； 3. 检查润磨机前给料胶带机的入带溜槽(4301-CHE-008/009)是否正常； 4. 4.检查润磨机润滑油站冷却水进口闸阀(65GA)是否正常。
8	Inspect Materials Level 检查物料料位	1. Check whether the lubricating oil of the lubricating oil station is sufficient, Whether the lubricating oil reflux condition is normal; 2. Check whether the material is stored in the grinding mill (4301-MIL-001/002); 3. Check the liquid level of the washing tower to ensure adequate circulating solution (4301-TOW-001); 4. Check whether the material is stored on the tape machine (4301-BCV-003), and there is no adhesive

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		deposit on the surface of the head and tail wheel and the belt joint. 1. 检查润滑油站润滑油是否足够，润滑油回流状况是否正常； 2. 检查润磨机内(4301-MIL-001/002)是否存料； 3. 检查洗涤塔液位，确保在循环溶液充足(4301-TOW-001)； 4. 4.检查胶带机(4301-BCV-003)上是否存料，首尾轮表面与皮带结合部无黏附积矿存在。
9	Start-Up Of the Moistening Mill 润磨机岗位开机	1. No-load start should be guaranteed when starting; 2. The driving sequence is along the material direction, from back to front that is the reverse drive, Each equipment needs to be switched on later;At the same time, confirm that the process behind the grinder post has been started and has the feeding conditions: Open the scrubber (4301-TOW-001) filling valve (LV001,50GA), load the tank into 3 / 4 full water, then close the filling valve (50GA), open the circulating pump for 3 minutes, and then release the outlet valve (80BF, 50BF) from the water in the scrubber ⟹ Close the outlet valve (80BF, 50BF) for wastewater discharge, open the water fill valve (50GA), and inject fresh water into the circulating tank until full ⟹ Close the water valve (50GA), open the water valve to adjust the height of the floating ball switch, and do not adjust arbitrarily after adjustment ⟹ Open the valve (80BF) and open the circulation pump (4301-PSO-001/002) ⟹ Open the induced draft fan (4301-FAN-001), The fan runs normally without load ⟹ Open the adhesive tape machine (4301-BCV-003) ⟹ Open grinding mill (4301-MIL-001/002) After checking the oil and oil level of the lubricating oil station meet the starting requirements, Open the cooling water of the lubricating oil station, open the high pressure and low pressure oil pump, top the mill shaft bush and lubricate the shaft bush, and observe the lubricating oil backflow in the oil return pipeline ⟹ After stopping for a long time, first open the slow drive motor to drive the mill slowly until the aster in the mill cylinder is loose and not bonded, stop the slow drive motor, and remove the slow drive reducer ⟹ Minimum frequency frequency conversion start grinder ⟹ Feed tape machine before opening

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№	Stage	Key Points
		the grinding mill. 1. 启动时应保证空载启动； 2. 开车顺序为沿物料走向，从后往前开即逆向开车，各设备都需要延时开启；同时，确认润磨机岗位后面的工序已经开车并具备给料条件： 开启洗涤塔(4301-TOW-001)补水阀(LV001,50GA)，将水箱装入3/4满清水，然后关闭补水阀(50GA)，开启循环水泵3分钟，然后将废水排放的出口阀(80BF,50BF)释放洗涤塔中之清水→关闭废水排放的出口阀(80BF,50BF)，开启补水阀(50GA)，在循环水槽内再次注入清水直到满为止→关闭补水阀(50GA)，打开补水阀调整浮球开关高度，调整好后不得任意再调整→打开阀门(80BF)，开启循环泵(4301-PSO-001/002)→开启引风机(4301-FAN-001)，待风机无负荷运转正常→开启胶带机(4301-BCV-003)→开启润磨机(4301-MIL-001/002)检查润滑油站油温和油位符合开机要求后，开启润滑油站冷却水，开启高压和低压油泵，对润磨机轴瓦进行顶起并对轴瓦进行润滑等，观察回油管路有润滑油回流→长时间停车状态再开车，首先开启慢驱动电机对润磨机进行慢速驱动，直到润磨机筒体内的焙砂松动没有粘结，停止慢驱电机，并将慢驱减速机脱开→最低频率变频启动润磨机→开启润磨机前给料胶带机。
10	The Moistening Mill Stops 润磨机岗位停机	1. Orderly shutdown a. The materials in the standby machine should be discharged before stopping; b. Stop in reverse order of driving order , Each equipment needs to stop; Stop the feed tape machine before the mill, Stop the mill (4301-MIL-001/002) tape machine and stop the mill, Close the high-pressure pump of the lubricating oil station, Close the cooling water of the lubricating oil station→Stop the tape machine (4301-BCV-003)→Stop the fan (4301-FAN-001)→ Stop the circulating pump (4301-PSO-001/002) and close the valve (80BF,LIT001)→Stop scrubber

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№	Stage	Key Points
		(4301-TOW-001)⟹Clean the site for hygiene. 1. 正常停车 a. 停车时应待机内物料排净后再停车； b. 按照开车顺序逆序停车顺序，各设备都需要延时停止： 停止润磨机前给料胶带机，停止润磨机(4301-MIL-001/002)胶带机，停止润磨机，关闭润滑油站高压泵，关闭润滑油站冷却水⟹停止胶带机(4301-BCV-003)⟹停止风机(4301-FAN-001)⟹停止循环泵(4301-PSO-001/002)，泵停稳后关闭阀门(80BF,LIT001)⟹停止洗涤塔(4301-TOW-001)⟹打扫现场卫生。 2. Emergency Stop a. In an emergency, close Minding before feeding tape machine, the grinding machine (4301-MIL-001/002), tape machine (4301-BCV-003), fan (4301-FAN-001), circulating pump (4301-PSO-001/002) after all stop operation, check whether the start/stop button of each equipment is in stop state, close the circulating pump outlet valve (80BF) , the upper level equipment shall stop; b. After the accident, the equipment is restored to normal. If production still needs to be conducted, and the operation shall be restarted in the driving sequence; c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，关闭润磨机前给料胶带机，润磨机(4301-MIL-001/002)、胶带机(4301-BCV-003)、风机(4301-FAN-001)、循环泵(4301-PSO-001/002)，全部停止运行后，检查各设备的启动/停止按钮是否处于停止状态，关闭循环泵出口阀门 (80BF)，同时，上一级设备均需停车； b. 事故处理后，设备恢复正常，如仍需要进行生产，按照开车顺序重新启动操作；

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№	Stage	Key Points
		c. 事故处理后，设备恢复正常，如不需要进行生产，按照正常停车顺序进行操作。
11	Permits ＆Isolation Sheets 许可证和隔离表	1. Only authorised isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

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8. Moistening Mill Post 润磨机岗位 Questionnaire

Declaration of Understanding

I (_____ Learner _____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____ Assessor _____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

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9. Appendix

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Standard Operating Procedure
标准操作程序

Moistening mill post in the chloride plant
CP 润磨机岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
Date Approved 审批日期:	Date Issued 颁布日期:	Next Review Date 下次审核日期:
Details of Revision 修订明细:		

Revision Details 修订明细

Rev. No 版本号	Date 日期	Section 章节	Page 页码	Revision Description 修订说明	Reviewed By 审核人
1					
2					
3					

Approvals 批准

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